

FOSS Security Tools: The *arp-scan* Package

Free and open source software (FOSS) offers a range of security tools. In this first article in the new series on FOSS security tools, we will explore the *arp-scan* package.



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arp-scan is a command-line utility for discovering IPv4 hosts in the network. It is written by Roy Hills and is available for GNU/Linux, MacOS, BSD and Solaris. It is primarily written in C and released under the GPLv3 licence. You can use it to send ARP (address resolution protocol) packets with configurable options, and customise the fields in the ARP packet. The tool can decode the packets received from the hosts and provide vendor information based on the MAC address. It can also fingerprint the hosts on the local network. You can install the same on Ubuntu using:

```
$ sudo apt install arp-scan
```

The version of *arp-scan* that we will use is 1.9.7 on Ubuntu 22.04.4 LTS:

```
$ arp-scan --version
arp-scan 1.9.7
```

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```
libpcap version 1.10.1 (with TPACKET_V3)
```

Localnet

The `--localnet` or `-l` option generates the addresses from the interface configuration and netmask, and scans them in the local network. For example:

```
$ sudo arp-scan --localnet
```

```
Interface: eth0, type: EN10MB, MAC: d4:ba:ba:af:ff:f1, IPv4:
10.0.0.104
Starting arp-scan 1.9.7 with 256 hosts (https://github.com/
royhills/arp-scan)
10.0.0.1 2c:54:91:88:c9:e3 (Unknown)
10.0.0.100 8c:1f:64:a2:d0:01 IEEE Registration Authority
10.0.0.103 00:0a:95:9d:68:16 (Unknown: locally administered)
10.0.0.1 2c:54:91:88:c9:e3 (Unknown) (DUP: 2)
10.0.0.1 2c:54:91:88:c9:e3 (Unknown) (DUP: 3)
```

```
5 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.9.7: 256 hosts scanned in 1.957 seconds
(130.81 hosts/sec). 5 responded
```

Verbose

You can increase the verbosity of the output messages by increasing the number of v's in the option.

```
$ sudo arp-scan -vv 10.0.0.100
Interface: eth0, type: EN10MB, MAC: d4:ba:ba:af:ff:f1, IPv4:
10.0.0.103
DEBUG: pcap filter string: "ether dst d4:ba:ba:af:ff:f1 and
```